

Crack The Case - Play Button Decoder

NxtWave Academy Workshop & Case Study Submission

TRACK: NETFLIX — THE DISCOVERY PROBLEM

PROJECT TITLE	Cyber Terminal - Interactive Play Button Vault
SELECTED TRACK	NETFLIX — The Discovery Problem
TECH STACK USED	HTML5, Tailwind CSS, Vanilla JavaScript, Canvas Confetti API
ARCHITECTURE / DEPLOYMENT	Client-Side Web App (Netlify / GitHub Pages Ready)

1. Project Description & Core Concept

This project is a high-engagement, dynamic web application designed to solve content discovery friction by gamifying the credential verification process. Built specifically for the NxtWave "Crack The Case" Play Button Challenge, the application presents users with an interactive cyber terminal.

Instead of static text submissions, users solve multi-stage puzzle ciphers derived from workshop session clues. Successfully cracking all stages unlocks a custom-rendered **Digital Gold Play Button Award** complete with confetti visual effects and a downloadable verified completion badge.

2. Key Features & Functionality

FEATURE MODULE	TECHNICAL IMPLEMENTATION	USER IMPACT
Progressive Unlocking	Dynamic state machine (33% → 66% → 100%) in JS	Guides user step-by-step through discovery stages.
Cyber Terminal Logs	Real-time DOM appending with timestamps	Provides visual backend execution logs and system transparency.
Interactive Validation	Case-insensitive array matching & error triggers	Instant feedback for correct and incorrect clue entries.
Digital Certificate Export	Native browser print CSS integration	Allows user to print or save a verified PDF accomplishment badge.

3. Input / Output Specifications

- **Detective Name (Input):** User's full name (e.g., *Sai Kiran*) — personalizes the final award output.
- **Student ID / Email (Input):** Identifier (e.g., *NXT-2026-CS*) — attached to verified terminal logs.
- **Clue 1 Key:** NXTWAVE or PLAYBUTTON — advances progress to 66%.
- **Clue 2 Key:** 1024 — advances progress to 100%.

- **Clue 3 Key:** SECRET or SOLVED — opens the Gold Vault screen.
- **Final Output:** Interactive Gold Play Button Display Card, Confetti Burst, Timestamped Security Hash, and Printable Badge.

4. Alignment with "NETFLIX — The Discovery Problem"

In digital media and streaming platforms, driving user engagement through content discovery is a major challenge. This web app addresses "The Discovery Problem" by replacing passive viewing with active puzzle-solving mechanics. By turning content retrieval into a interactive quest, user retention and completion rates increase significantly.

5. Core Logic Architecture (JavaScript Event Handler)

```
// Verification Engine & Terminal Logger Logic
function processClue(e) {
  e.preventDefault();
  const inputVal = document.getElementById('clue-input').value.trim().toUpperCase();
  const activeStageData = stages[currentStage - 1];

  if (activeStageData.key.includes(inputVal)) {
    logToTerminal(`[VERIFIED]: Stage ${currentStage} unlocked.`, 'success');
    if (currentStage < 3) {
      currentStage++;
      updateStageUI();
    } else {
      finishCase(); // Triggers Confetti & Gold Award Card
    }
  } else {
    logToTerminal(`[FAILED]: Invalid key for Stage ${currentStage}.`, 'error');
  }
}
```